Unit 4 Project Brief

Flight Performance Data Investigation

Flight Performance: Statistics, Kinematics & Test Data

Challenge Statement: Design and conduct a controlled flight or motion investigation, collect repeated data, analyze variation, and use kinematic reasoning to explain performance.

Project Overview

In this investigation, your team uses repeated trials and motion data to evaluate how an aerospace object moves or performs. The focus is not just making something fly or move - it is using statistics, graphs, and kinematics to make a defensible performance claim.

Mission Scenario

Aerospace engineers rely on test data before trusting a system. Gliders, projectiles, drones, rovers, launch devices, and dropped objects all produce motion data that can be measured, graphed, compared, and used to improve design decisions. Your team will run a controlled investigation and defend a conclusion using evidence.

Design Requirements

- Select an approved investigation such as glider flight, projectile launch, drop test, drone route timing, rover motion, or launch angle/range study.
- Identify independent, dependent, and controlled variables.
- Collect repeated trial data under consistent testing conditions.
- Create at least one graph or visual representation of the data.
- Calculate appropriate statistics such as mean, median, range, standard deviation, percent error, success rate, or variation between trials.
- Use kinematic ideas such as distance, displacement, velocity, acceleration, free fall, projectile motion, range, or launch angle to explain results.

Constraints

- Test setup must be safe, repeatable, and approved before trials begin.
- Objects must not be launched, dropped, or flown toward people, fragile objects, or unsafe areas.
- Data must come from multiple trials, not a single best run.
- Only one primary variable should be changed at a time unless approved.
- Final claim must be based on data and calculations, not opinion or appearance.

Required Engineering Evidence

Evidence Type	What to include
Investigation Plan	Research question, variables, procedure, equipment list, and safety notes.
Raw Data	Repeated trial data with units, notes, and any rejected-trial explanation.
Statistics	Mean, variation, range, standard deviation, success rate, or other assigned data analysis.
Kinematics	Motion calculations or graphs connecting performance to physics concepts.

Conclusion	Evidence-based claim about performance and a recommended improvement or next test.

Project Checkpoints

1. Research question approved
2. Test plan approved
3. Trial data collected
4. Graphs/statistics complete
5. Final performance claim submitted

Success Criteria

Category	Evidence of success
Experimental Design	Variables, controls, and procedure create a fair and repeatable investigation.
Data Quality	Repeated trials are complete, accurate, and organized with units.
Analysis	Statistics, graphs, and kinematics are correct and meaningful.
Conclusion	Final claim is specific, evidence-based, and connected to design or performance decisions.

Final Submission

Submit a complete design package that includes your notebook evidence, sketches or CAD evidence, prototype/build evidence, test data, calculations or analysis, and a final design review presentation. Your final recommendation should clearly explain what your design does, how well it performs, and what should be improved next.